

Problem Statement

Clinical Operations — Business Intelligence Dashboard Project

1. Background

Healthcare organizations rely on efficient scheduling and provider utilization to deliver quality patient care while controlling operating costs. Clinical operations teams within hospital systems track appointment volume, no-show rates, patient wait times, and departmental performance to identify where capacity is being lost and where the patient experience can be improved.

This project simulates a clinical operations analytics environment for a multi-department outpatient system, covering patient scheduling, provider assignments, and appointment outcomes across six clinical departments (Cardiology, Orthopedics, Pediatrics, Oncology, Dermatology, and Neurology).

2. Problem Statement

Clinical leadership currently lacks a centralized, reliable view of appointment performance across departments and providers. Appointment, patient, and provider data exists in raw operational form with no standardized reporting layer, making it difficult to answer basic operational questions such as:

- How much scheduled capacity is lost to no-shows and cancellations?
- Which departments and providers have the highest no-show rates?
- Where are patient wait times longest, and is that concentrated with specific providers?
- Can leadership trust the underlying data, or are there quality issues (duplicates, missing records, invalid references) distorting the numbers?

Without a clean, modeled, and visualized dataset, these questions require manual, ad-hoc analysis each time they're asked — an unsustainable and error-prone process for a growing clinical operation.

3. Objectives

- Consolidate patient, provider, department, and appointment data into a single cloud data warehouse (Snowflake).
- Conduct a formal data quality audit to identify duplicates, orphaned records, invalid values, and missing data.
- Build a cleaned staging layer in SQL that resolves identified data quality issues without breaking referential integrity.
- Design a star-schema data model in Power BI connecting appointments (fact) to patients, providers, and departments (dimensions).
- Develop DAX measures covering appointment volume, no-show rate, wait time, and data quality metrics.
- Deliver an interactive Power BI dashboard answering leadership's key operational questions, with clear findings and recommendations.

4. Scope

In Scope

- Appointment records across 6 departments and 25 providers, ~420 patients, ~3,000 appointments
- Data quality audit and remediation via SQL in Snowflake
- Star-schema data model and DAX measures in Power BI
- Three-page dashboard: Executive Summary, Operational Drill-Down, Data Quality Audit

Out of Scope

- Real-time / streaming data integration (DirectQuery)
- Clinical outcomes or diagnosis-level data
- Billing, claims, or revenue cycle data

5. Data Sources

Synthetic data generated to mirror real-world clinical operations structure and data quality issues, consisting of four tables: patients, providers, departments, and appointments, loaded into Snowflake as raw CSV files.

6. Tools & Technologies

- Snowflake — cloud data warehouse, staging & transformation layer (SQL)
- Power BI — data modeling (star schema) and dashboard visualization
- DAX — measures for KPIs and time intelligence

7. Expected Deliverables

- A cleaned, modeled dataset in Snowflake (RAW and STAGING schemas)
- A documented data quality audit with before/after metrics
- A 3-page interactive Power BI dashboard
- A final report summarizing methodology, findings, and recommendations